



Dhirubhai Ambani Institute of Information & Communication Technology
Mid Semester Test-2, 2nd Semester 2023- 24

Course Title IT694 Computer Networks

Date 21 March 2024

OPEN NOTES

Max Marks

15

Time

1 Hour

Be brief in your answers. All questions carry 5 marks each.

1. TCP computes the time-out T_{out} for each packet as a function of estimated $\langle RTT \rangle$ and variance of RTT.
 - (a) Write down the expression used by the TCP for $\langle RTT \rangle$ and T_{out} .
 - (b) What is a good estimate of the T_{out} for the first segment sent by the TCP.
 - (c) Qualitatively explain the impact of the parameter α , for large and small values.
 - (d) What is the standard formula for the Variance of a set of values? Why does TCP use a slightly different formula?

2. Stop and Wait (SnW) protocol uses sequence numbers (SEQ), time outs (T_o), and acknowledgement numbers (ACK) to achieve reliable transmission.
 - (a) Under what conditions, does the efficiency of the SnW protocol become quite low?
 - (b) If the packet error probability is P , the average number of retransmissions is $1/P$. Without deriving, can you suggest the expression for efficiency in the presence of errors.
 - (c) Instead of sending an ACK for packet received correctly, a receiver may be programmed to send a negative acknowledgement (NACK) when an incorrect packet is received. Design the variant of SnW that uses NACK instead of ACK.
 - (d) Draw the FSM diagram for the receiver and sender for this NACK based SnW.

3. In the context of Go-back-N (GBN) and Selective Repeat (SR) protocols,
 - (a) Define the terms Pipelining and Piggybacking. How do these help improve the performance of the link reliability protocols?
 - (b) Window size and the SEQ number size must satisfy a relation for these protocols to work correctly. Give these relations and give an example for both GBN and SR to show that these protocols don't work correctly if the said relation is not satisfied.
 - (c) Without deriving a formula, explain why the efficiency of these protocols is 1 when noise is not present.